Precalculus — Algebra I

Version 4 $- \epsilon$

by

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$\sigma o \phi \acute{\imath} \alpha$

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Sophia Publishing
CHENNAI
https://sophia-publishing.github.io/

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Preface

Thank you for your interest in our book, but more importantly, thank you for taking the time to read the Preface. I always read the Prefaces of the textbooks which I use in my classes because I believe it is in the Preface where I begin to understand the authors - who they are, what their motivation for writing the book was, and what they hope the reader will get out of reading the text. Pedagogical issues such as content organization and how professors and students should best use a book can usually be gleaned out of its Table of Contents, but the reasons behind the choices authors make should be shared in the Preface. Also, I feel that the Preface of a textbook should demonstrate the authors' love of their discipline and passion for teaching, so that I come away believing that they really want to help students and not just make money. Thus, I thank my fellow Preface-readers again for giving me the opportunity to share with you the need and vision which guided the creation of this book and passion which both Carl and I hold for Mathematics and the teaching of it.

Carl and I are natives of Northeast Ohio. We met in graduate school at Kent State University in 1997. I finished my Ph.D in Pure Mathematics in August 1998 and started teaching at Lorain County Community College in Elyria, Ohio just two days after graduation. Carl earned his Ph.D in Pure Mathematics in August 2000 and started teaching at Lakeland Community College in Kirtland, Ohio that same month. Our schools are fairly similar in size and mission and each serves a similar population of students. The students range in age from about 16 (Ohio has a Post-Secondary Enrollment Option program which allows high school students to take college

courses for free while still in high school.) to over 65. Many of the "non-traditional" students are returning to school in order to change careers. A majority of the students at both schools receive some sort of financial aid, be it scholarships from the schools' foundations, state-funded grants or federal financial aid like student loans, and many of them have lives busied by family and job demands. Some will be taking their Associate degrees and entering (or re-entering) the workforce while others will be continuing on to a four-year college or university. Despite their many differences, our students share one common attribute: they do not want to spend $200 on a College Algebra book.

The challenge of reducing the cost of textbooks is one that many states, including Ohio, are taking quite seriously. Indeed, state-level leaders have started to work with faculty from several of the colleges and universities in Ohio and with the major publishers as well. That process will take considerable time so Carl and I came up with a plan of our own. We decided that the best way to help our students right now was to write our own College Algebra book and give it away electronically for free. We were granted sabbaticals from our respective institutions for the Spring semester of 2009 and actually began writing the textbook on December 16, 2008. Using an open-source text editor called TexNicCenter and an open-source distribution of LaTeX called MikTex 2.7, Carl and I wrote and edited all of the text, exercises and answers and created all of the graphs (using Metapost within LaTeX) for Version $0.\overline{9}$ in about eight months. (We choose to create a text in only black and white to keep printing costs to a minimum for those students who prefer a printed edition. This somewhat Spartan page layout stands in sharp relief to the explosion of colors found in most other College Algebra texts, but neither Carl nor I believe the four-color print adds anything of value.) I used the book in three sections of College Algebra at Lorain County Community College in the Fall of 2009 and Carl's colleague, Dr. Bill Previts, taught a section of College Algebra at Lakeland with the book that semester as well. Students had the option of downloading the book as a .pdf file from our website www.stitz-zeager.com[1] or buying a low-cost printed version from our colleges' respective bookstores. (By

[1] http://www.stitz-zeager.com

giving this book away for free electronically, we end the cycle of new editions appearing every 18 months to curtail the used book market.) During Thanksgiving break in November 2009, many additional exercises written by Dr. Previts were added and the typographical errors found by our students and others were corrected. On December 10, 2009, Version $\sqrt{2}$ was released. The book remains free for download at our website and by using Lulu.com[2] as an on-demand printing service, our bookstores are now able to provide a printed edition for just under $19. Neither Carl nor I have, or will ever, receive any royalties from the printed editions. As a contribution back to the open-source community, all of the LaTeX files used to compile the book are available for free under a Creative Commons License on our website as well. That way, anyone who would like to rearrange or edit the content for their classes can do so as long as it remains free.

The only disadvantage to not working for a publisher is that we don't have a paid editorial staff. What we have instead, beyond ourselves, is friends, colleagues and unknown people in the open-source community who alert us to errors they find as they read the textbook. What we gain in not having to report to a publisher so dramatically outweighs the lack of the paid staff that we have turned down every offer to publish our book. (As of the writing of this Preface, we've had three offers.) By maintaining this book by ourselves, Carl and I retain all creative control and keep the book our own. We control the organization, depth and rigor of the content which means we can resist the pressure to diminish the rigor and homogenize the content so as to appeal to a mass market. A casual glance through the Table of Contents of most of the major publishers' College Algebra books reveals nearly isomorphic content in both order and depth. Our Table of Contents shows a different approach, one that might be labeled "Functions First." To truly use The Rule of Four, that is, in order to discuss each new concept algebraically, graphically, numerically and verbally, it seems completely obvious to us that one would need to introduce functions first. (Take a moment and compare our ordering to the classic "equations first, then the Cartesian Plane and THEN functions" approach seen in most of

[2] http://www.lulu.com/content/paperback-book/college-algebra/7513097

the major players.) We then introduce a class of functions and discuss the equations, inequalities (with a heavy emphasis on sign diagrams) and applications which involve functions in that class. The material is presented at a level that definitely prepares a student for Calculus while giving them relevant Mathematics which can be used in other classes as well. Graphing calculators are used sparingly and only as a tool to enhance the Mathematics, not to replace it. The answers to nearly all of the computational homework exercises are given in the text and we have gone to great lengths to write some very thought provoking discussion questions whose answers are not given. One will notice that our exercise sets are much shorter than the traditional sets of nearly 100 "drill and kill" questions which build skill devoid of understanding. Our experience has been that students can do about 15-20 homework exercises a night so we very carefully chose smaller sets of questions which cover all of the necessary skills and get the students thinking more deeply about the Mathematics involved.

Critics of the Open Educational Resource movement might quip that "open-source is where bad content goes to die," to which I say this: take a serious look at what we offer our students. Look through a few sections to see if what we've written is bad content in your opinion. I see this open-source book not as something which is "free and worth every penny", but rather, as a high quality alternative to the business as usual of the textbook industry and I hope that you agree. If you have any comments, questions or concerns please feel free to contact me at jeff@stitz-zeager.com or Carl at carl@stitz-zeager.com.

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January 25, 2010

Chapter 1

Introduction to Functions

1.1 Functions and their Representations

1.2 Constant and Linear Functions

1.3 Absolute Value Functions

1.4 Quadratic Functions

Chapter 2

Polynomial Functions

2.1 Graphs of Polynomial Functions

2.2 The Remainder and Factor Theorems

2.3 Real Zeros of Polynomials

2.4 Complex Zeros and the Fundamental Theorem of Algebra

Chapter 3

Rational Functions

3.1 Introduction to Rational Functions

3.2 Graphs of Rational Functions

3.3 Inequalities involving Rational Functions and Applications

Chapter 4

Root, Radical and Power Functions

4.1 Root and Radical Functions

4.2 Power Functions

4.3 Equations and Inequalities involving Power Functions

Chapter 5

Further Topics on Functions

5.1 Graphs of Functions

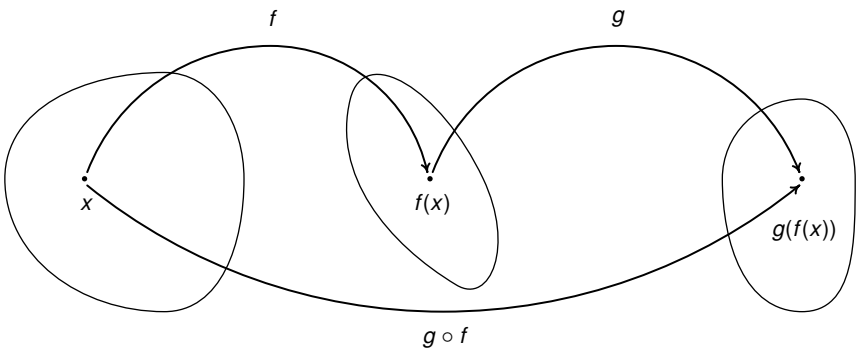
5.2 Function Arithmetic

5.3 Function Composition

In Section **??**, we saw how the arithmetic of real numbers carried over into an arithmetic of functions. In this section, we discuss another way to combine functions which is unique to functions and isn't shared with real numbers - function **composition**.

Definition 5.3.1. Let f and g be functions where the real number x is in the domain of f and the real number $f(x)$ is in the domain of g. The **composite** of g with f, denoted $g \circ f$, and read 'g composed with f' is defined by the formula: $(g \circ f)(x) = g(f(x))$.

To compute $(g \circ f)(x)$, we use the formula given in Defintion 5.3.1: $(g \circ f)(x) = g(f(x))$. However, from a procedural viewpoint, Defintion 5.3.1 tells us the output from $g \circ f$ is found by taking the output from f, $f(x)$, and then making that the input to g. From this perspective, we see $g \circ f$ as a two step process taking an input x and first applying the procedure f then applying the procedure g. Abstractly, we have



In the expression $g(f(x))$, the function f is often called the 'inside' function while g is often called the 'outside' function. When evaluating composite function values we present two methods in the example below: the 'inside out' and 'outside in' methods.

Example 5.3.1. Let $f(x) = x^2 - 4x$, $g(t) = 2 - \sqrt{t+3}$, and $h(s) = \dfrac{2s}{s+1}$.

In numbers 1 - 3, find the indicated function value.

1. $(g \circ f)(1)$
2. $(f \circ g)(1)$
3. $(g \circ g)(6)$

In numbers 4 - 10, find and simplify the indicated composite functions. State the domain of each.

4. $(g \circ f)(x)$
5. $(f \circ g)(t)$
6. $(g \circ h)(s)$
7. $(h \circ g)(t)$
8. $(h \circ h)(x)$
9. $(h \circ (g \circ f))(x)$
10. $((h \circ g) \circ f)(x)$

Solution.

1. Using Definition 5.3.1, $(g \circ f)(1) = g(f(1))$. Since $f(1) = (1)^2 - 4(1) = -3$ and $g(-3) = 2 - \sqrt{(-3)+3} = 2$, we have $(g \circ f)(1) = g(f(1)) = g(-3) = 2$.

2. By definition, $(f \circ g)(1) = f(g(1))$. We find $g(1) = 2 - \sqrt{1+3} = 0$, and $f(0) = (0)^2 - 4(0) = 0$, so $(f \circ g)(1) = f(g(1)) = f(0) = 0$. Comparing this with our answer to the last problem, we see that $(g \circ f)(1) \neq (f \circ g)(1)$ which tells us function composition is not commutative,[1]

3. Since $(g \circ g)(6) = g(g(6))$, we 'iterate' the process g: that is, we apply the process g to 6, then apply the process g again. We find $g(6) = 2 - \sqrt{6+3} = -1$, and $g(-1) = 2 - \sqrt{(-1)+3} = 2 - \sqrt{2}$, so $(g \circ g)(6) = g(g(6)) = g(-1) = 2 - \sqrt{2}$.

4. By definition, $(g \circ f)(x) = g(f(x))$. We now illustrate *two* ways to approach this problem.

   - *inside out*: We substitute $f(x) = x^2 - 4x$ in for t in the expression $g(t)$ and get

$$\begin{aligned}(g \circ f)(x) = g(f(x)) &= g\left(x^2 - 4x\right) \\ &= 2 - \sqrt{\left(x^2 - 4x\right) + 3} \\ &= 2 - \sqrt{x^2 - 4x + 3}\end{aligned}$$

[1] That is, in general, $g \circ f \neq f \circ g$. This shouldn't be too surprising, since, in general, the order of processes matters: adding eggs to a cake batter then baking the cake batter has a much different outcome than baking the cake batter then adding eggs.

Hence, $(g \circ f)(x) = 2 - \sqrt{x^2 - 4x + 3}$.

- *outside in*: We use the formula for g first to get

$$\begin{aligned}(g \circ f)(x) = g(f(x)) &= 2 - \sqrt{f(x) + 3} \\ &= 2 - \sqrt{\left(x^2 - 4x\right) + 3} \\ &= 2 - \sqrt{x^2 - 4x + 3}\end{aligned}$$

We get the same answer as before, $(g \circ f)(x) = 2 - \sqrt{x^2 - 4x + 3}$.

To find the domain of $g \circ f$, we need to find the elements in the domain of f whose outputs $f(x)$ are in the domain of g. Since the domain of f is all real numbers, we focus on finding the range elements compatible with g. Owing to the presence of the square root in the formula $g(t) = 2 - \sqrt{t + 3}$ we require $t \geq -3$. Hence, we need $f(x) \geq -3$ or $x^2 - 4x \geq -3$. To solve this inequality we rewrite as $x^2 - 4x + 3 \geq 0$ and use a sign diagram. Letting $r(x) = x^2 - 4x + 3$, we find the zeros of r to be $x = 1$ and $x = 3$ and obtain

$$\begin{array}{ccccc} (+) & 0 & (-) & 0 & (+) \\ \hline & 1 & & 3 & \end{array}$$

Our solution to $x^2 - 4x + 3 \geq 0$, and hence the domain of $g \circ f$, is $(-\infty, 1] \cup [3, \infty)$.

5. To find $(f \circ g)(t)$, we find $f(g(t))$.

- *inside out*: We substitute the expression $g(t) = 2 - \sqrt{t + 3}$ in for x in the formula $f(x)$ and get

$$\begin{aligned}(f \circ g)(t) &= f(g(t)) \\ &= f\left(2 - \sqrt{t + 3}\right) \\ &= \left(2 - \sqrt{t + 3}\right)^2 - 4\left(2 - \sqrt{t + 3}\right) \\ &= 4 - 4\sqrt{t + 3} + \left(\sqrt{t + 3}\right)^2 - 8 + 4\sqrt{t + 3} \\ &= 4 + t + 3 - 8\end{aligned}$$

$$= t - 1$$

- *outside in*: We use the formula for $f(x)$ first to get

$$\begin{aligned}(f \circ g)(t) &= f(g(t)) \\ &= (g(t))^2 - 4\,(g(t)) \\ &= \left(2 - \sqrt{t+3}\right)^2 - 4\left(2 - \sqrt{t+3}\right) \\ &= t - 1 \qquad \text{(same algebra as before)}\end{aligned}$$

Thus we get $(f \circ g)(t) = t - 1$. To find the domain of $f \circ g$, we look for the elements t in the domain of g whose outputs, $g(t)$ are in the domain of f. As mentioned previously, the domain of g is limited by the presence of the square root to $\{t \in \mathbb{R} \mid t \geq -3\}$ while the domain of f is all real numbers. Hence, the domain of $f \circ g$ is restricted only by the domain of g and is $\{t \in \mathbb{R} \mid t \geq -3\}$ or, using interval notation, $[-3, \infty)$. Note that as with Example **??** in Section **??**, had we used the simplified formula for $(f \circ g)(t) = t - 1$ to determine domain, we would have arrived at the incorrect answer.

6. To find $(g \circ h)(s)$, we compute $g(h(s))$.

   - *inside out*: We substitute $h(s)$ in for t in the expression $g(t)$ to get

$$\begin{aligned}(g \circ h)(s) &= g(h(s)) \\ &= g\left(\frac{2s}{s+1}\right) \\ &= 2 - \sqrt{\left(\frac{2s}{s+1}\right) + 3} \\ &= 2 - \sqrt{\frac{2s}{s+1} + \frac{3(s+1)}{s+1}} \\ &\qquad \text{(get common denominators)}\end{aligned}$$

$$= 2 - \sqrt{\frac{5s+3}{s+1}}$$

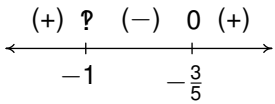
- *outside in*: We use the formula for $g(t)$ first to get

$$\begin{aligned}(g \circ h)(s) &= g(h(s)) \\ &= 2 - \sqrt{h(s)+3} \\ &= 2 - \sqrt{\left(\frac{2s}{s+1}\right)+3} \\ &= 2 - \sqrt{\frac{5s+3}{s+1}} \\ &\text{(get common denominators as before)}\end{aligned}$$

To find the domain of $g \circ h$, we need the elements in the domain of h so that $h(s)$ is in the domain of g. Owing to the $s+1$ in the denominator of the expression $h(s)$, we require $s \neq -1$. Once again, because of the square root in $g(t) = 2 - \sqrt{t+3}$, we need $t \geq -3$ or, in this case $h(s) \geq -3$. To use a sign diagram to solve, we rearrange this inequality:

$$\begin{aligned}\frac{2s}{s+1} &\geq -3 \\ \frac{2s}{s+1} + 3 &\geq 0 \\ \frac{5s+3}{s+1} &\geq 0 \qquad \text{(get common denominators as before)}\end{aligned}$$

Defining $r(s) = \frac{5s+3}{s+1}$, we see r is undefined at $s = -1$ (a carry over from the domain restriction of h) and $r(s) = 0$ at $s = -\frac{3}{5}$. Our sign diagram is



hence our domain is $(-\infty, -1) \cup \left[-\frac{3}{5}, \infty\right)$.

7. We find $(h \circ g)(t)$ by finding $h(g(t))$.

   - *inside out*: We substitute the expression $g(t)$ for s in the formula $h(s)$

$$\begin{aligned}(h \circ g)(t) &= h(g(t)) = h\left(2 - \sqrt{t+3}\right) \\ &= \frac{2\left(2 - \sqrt{t+3}\right)}{\left(2 - \sqrt{t+3}\right) + 1} \\ &= \frac{4 - 2\sqrt{t+3}}{3 - \sqrt{t+3}}\end{aligned}$$

   - *outside in*: We use the formula for $h(s)$ first to get

$$\begin{aligned}(h \circ g)(t) &= h(g(t)) = \frac{2\,(g(t))}{(g(t)) + 1} \\ &= \frac{2\left(2 - \sqrt{t+3}\right)}{\left(2 - \sqrt{t+3}\right) + 1} \\ &= \frac{4 - 2\sqrt{t+3}}{3 - \sqrt{t+3}}\end{aligned}$$

   To find the domain of $h \circ g$, we need the elements of the domain of g so that $g(t)$ is in the domain of h. As we've seen already, for t to be in the domain of g, $t \geq -3$. For s to be in the domain of h, $s \neq -1$, so we require $g(t) \neq -1$. Hence, we solve $g(t) = 2 - \sqrt{t+3} = -1$ with the intent of excluding the solutions. Isolating the radical expression gives $\sqrt{t+3} = 3$ or $t = 6$. Sure enough, we check $g(6) = -1$ so we exclude $t = 6$ from the domain of $h \circ g$. Our final answer is $[-3, 6) \cup (6, \infty)$.

8. To find $(h \circ h)(s)$ we find $h(h(s))$:

   - *inside out*: We substitute the expression $h(s)$ for s in the expression $h(s)$ into h to get

$$
\begin{aligned}
(h \circ h)(s) &= h(h(s)) = h\left(\frac{2s}{s+1}\right) \\
&= \frac{2\left(\frac{2s}{s+1}\right)}{\left(\frac{2s}{s+1}\right)+1} \\
&= \frac{\frac{4s}{s+1}}{\frac{2s}{s+1}+1} \cdot \frac{(s+1)}{(s+1)} \\
&= \frac{\frac{4s}{s+1} \cdot (s+1)}{\left(\frac{2s}{s+1}\right) \cdot (s+1) + 1 \cdot (s+1)} \\
&= \frac{\frac{4s}{\cancel{(s+1)}} \cdot \cancel{(s+1)}}{\frac{2s}{\cancel{(s+1)}} \cdot \cancel{(s+1)} + s + 1} \\
&= \frac{4s}{3s+1}
\end{aligned}
$$

   - *outside in*: This approach yields

$$
\begin{aligned}
(h \circ h)(s) = h(h(s)) &= \frac{2(h(s))}{h(s)+1} \\
&= \frac{2\left(\frac{2s}{s+1}\right)}{\left(\frac{2s}{s+1}\right)+1}
\end{aligned}
$$

$$= \frac{4s}{3s+1} \qquad \text{(same algebra as before)}$$

To find the domain of $h \circ h$, we need to find the elements in the domain of h so that the outputs, $h(s)$ are also in the domain of h. The only domain restriction for h comes from the denominator: $s \neq -1$, so in addition to this, we also need $h(s) \neq -1$. To this end, we solve $h(s) = -1$ and exclude the answers. Solving $\frac{2s}{s+1} = -1$ gives $s = -\frac{1}{3}$. The domain of $h \circ h$ is $(-\infty, -1) \cup \left(-1, -\frac{1}{3}\right) \cup \left(-\frac{1}{3}, \infty\right)$.

9. The expression $(h \circ (g \circ f))(x)$ indicates that we first find the composite, $g \circ f$ and compose the function h with the result. We know from number 4 that $(g \circ f)(x) = 2 - \sqrt{x^2 - 4x + 3}$ with domain $(-\infty, 1] \cup [3, \infty)$. We now proceed as usual.

   - *inside out*: We substitute the expression $(g \circ f)(x)$ for s in the expression $h(s)$ first to get

$$\begin{aligned}
(h \circ (g \circ f))(x) &= h((g \circ f)(x)) = h\left(2 - \sqrt{x^2 - 4x + 3}\right) \\
&= \frac{2\left(2 - \sqrt{x^2 - 4x + 3}\right)}{\left(2 - \sqrt{x^2 - 4x + 3}\right) + 1} \\
&= \frac{4 - 2\sqrt{x^2 - 4x + 3}}{3 - \sqrt{x^2 - 4x + 3}}
\end{aligned}$$

   - *outside in*: We use the formula for $h(s)$ first to get

$$\begin{aligned}
(h \circ (g \circ f))(x) &= h((g \circ f)(x)) = \frac{2\left((g \circ f)(x)\right)}{\left((g \circ f)(x)\right) + 1} \\
&= \frac{2\left(2 - \sqrt{x^2 - 4x + 3}\right)}{\left(2 - \sqrt{x^2 - 4x + 3}\right) + 1} \\
&= \frac{4 - 2\sqrt{x^2 - 4x + 3}}{3 - \sqrt{x^2 - 4x + 3}}
\end{aligned}$$

To find the domain of $h \circ (g \circ f)$, we need the domain elements of $g \circ f$, $(-\infty, 1] \cup [3, \infty)$, so that $(g \circ f)(x)$ is in the domain of h. As we've seen several times already, the only domain restriction for h is $s \neq -1$, so we set $(g \circ f)(x) = 2 - \sqrt{x^2 - 4x + 3} = -1$ and exclude the solutions. We get $\sqrt{x^2 - 4x + 3} = 3$, and, after squaring both sides, we have $x^2 - 4x + 3 = 9$. We solve $x^2 - 4x - 6 = 0$ using the quadratic formula and obtain $x = 2 \pm \sqrt{10}$. The reader is encouraged to check that both of these numbers satisfy the original equation, $2 - \sqrt{x^2 - 4x + 3} = -1$ and also belong to the domain of $g \circ f$, $(-\infty, 1] \cup [3, \infty)$, and so must be excluded from our final answer.[2] Our final domain for $h \circ (f \circ g)$ is $(-\infty, 2 - \sqrt{10}) \cup (2 - \sqrt{10}, 1] \cup \left[3, 2 + \sqrt{10}\right) \cup \left(2 + \sqrt{10}, \infty\right)$.

10. The expression $((h \circ g) \circ f)(x)$ indicates that we first find the composite $h \circ g$ and then compose that with f. From number 7, we have

$$(h \circ g)(t) = \frac{4 - 2\sqrt{t+3}}{3 - \sqrt{t+3}}$$

with domain $[-3, 6) \cup (6, \infty)$.

- *inside out*: We substitute the expression $f(x)$ for t in the expression $(h \circ g)(t)$ to get

$$\begin{aligned}((h \circ g) \circ f)(x) &= (h \circ g)(f(x)) = (h \circ g)\left(x^2 - 4x\right) \\ &= \frac{4 - 2\sqrt{\left(x^2 - 4x\right) + 3}}{3 - \sqrt{\left(x^2 - 4x\right) + 3}} \\ &= \frac{4 - 2\sqrt{x^2 - 4x + 3}}{3 - \sqrt{x^2 - 4x + 3}}\end{aligned}$$

- *outside in*: We use the formula for $(h \circ g)(t)$ first to get

[2] We can approximate $\sqrt{10} \approx 3$ so $2 - \sqrt{10} \approx -1$ and $2 + \sqrt{10} \approx 5$.

$$\begin{aligned}((h \circ g) \circ f)(x) &= (h \circ g)(f(x)) = \frac{4 - 2\sqrt{(f(x)) + 3}}{3 - \sqrt{f(x)) + 3}} \\ &= \frac{4 - 2\sqrt{\left(x^2 - 4x\right) + 3}}{3 - \sqrt{\left(x^2 - 4x\right) + 3}} \\ &= \frac{4 - 2\sqrt{x^2 - 4x + 3}}{3 - \sqrt{x^2 - 4x + 3}}\end{aligned}$$

Since the domain of f is all real numbers, the challenge here to find the domain of $(h \circ g) \circ f$ is to determine the values $f(x)$ which are in the domain of $h \circ g$, $[-3, 6) \cup (6, \infty)$. At first glance, it appears as if we have two (or three!) inequalities to solve: $-3 \leq f(x) < 6$ and $f(x) > 6$. Alternatively, we could solve $f(x) = x^2 - 4x \geq -3$ and exclude the solutions to $f(x) = x^2 - 4x = 6$ which is not only easier from a procedural point of view, but also easier since we've already done both calculations. In number 4, we solved $x^2 - 4x \geq -3$ and obtained the solution $(-\infty, 1] \cup [3, \infty)$ and in number 9, we solved $x^2 - 4x - 6 = 0$ and obtained $x = 2 \pm \sqrt{10}$. Hence, the domain of $(h \circ g) \circ f$ is $(-\infty, 2 - \sqrt{10}) \cup (2 - \sqrt{10}, 1] \cup \left[3, 2 + \sqrt{10}\right) \cup \left(2 + \sqrt{10}, \infty\right)$. □

As previously mentioned, it should be clear from Example 5.3.1 that, in general, $g \circ f \neq f \circ g$, in other words, function composition is not *commutative*. However, numbers 9 and 10 demonstrate the **associative** property of function composition. That is, when composing three (or more) functions, as long as we keep the order the same, it doesn't matter which two functions we compose first. We summarize the important properties of function composition in the theorem below.

Theorem 5.3.1. Properties of Function Composition: Suppose f, g, and h are functions.

- **Associative Law of Composition:** $h \circ (g \circ f) = (h \circ g) \circ f$, provided the composite functions are defined.
- **Composition Identity:** The function $I(x) = x$ satisfies: $I \circ f = f \circ I = f$ for all functions, f.

By repeated applications of Definition 5.3.1, we find $(h \circ (g \circ f))(x) = h((g \circ f)(x)) = h(g(f(x)))$. Similarly, $((h \circ g) \circ f)(x) = (h \circ g)(f(x)) = h(g(f(x)))$. This establishes that the formulas for the two functions are the same. We leave it to the reader to think about why the domains of these two functions are identical, too. These two facts establish the equality $h \circ (g \circ f) = (h \circ g) \circ f$. A consequence of the associativity of function composition is that there is no need for parentheses when we write $h \circ g \circ f$. The second property can also be verified using Definition 5.3.1. Recall that the function $I(x) = x$ is called the *identity function* and was introduced in Exercise **??** in Section **??**. If we compose the function I with a function f, then we have $(I \circ f)(x) = I(f(x)) = f(x)$, and a similar computation shows $(f \circ I)(x) = f(I(x)) = f(x)$. This establishes that we have an identity for function composition much in the same way the function $I(x) = 1$ is an identity for function multiplication.

As we know, not all functions are described by formulas, and, moreover, not all functions are described by just *one* formula. The next example applies the concept of function composition to functions represented in various and sundry ways.

Example 5.3.2. Consider the following functions:

- $f(x) = 6x - x^2$
- $g(t) \begin{cases} 2t - 1 & \text{if } -1 \leq t < 3, \\ t^2 & \text{if } t \geq 3. \end{cases}$
- $h = \{(-3, 1), (-2, 6), (0, -2), (1, 5), (3, -1)\}$
- s whose graph is given in Figure 5.3.1:

1. Find and simplify the following function values:

   (a) $(g \circ f)(2)$ (b) $(h \circ g)(-1)$ (c) $(h \circ s)(-2)$

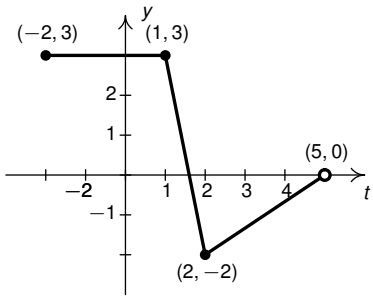


Figure 5.3.1: $y = s(t)$

(d) $(f \circ s)(0)$

2. Find and simplify a formula for $(g \circ f)(x)$.

3. Write $s \circ h$ as a set of ordered pairs.

Solution.

1. (a) To find $(g \circ f)(2) = g(f(2))$ we first find $f(2) = 6(2) - (2)^2 = 8$. Since $8 \geq 3$, we use the rule $g(t) = t^2$ so $g(8) = (8)^2 = 64$. Hence, $(g \circ f)(3) = g(f(3)) = g(8) = 64$.

   (b) Since $(h \circ g)(-1) = h(g(-1))$ we first need $g(-1)$. Since $-1 \leq -1 < 3$, we use the rule $g(t) = 2t - 1$ and find $g(-1) = 2(-1) - 1 = -3$. Next, we need $h(-3)$. Since $(-3, 1) \in h$, we have that $h(-3) = 1$. Putting it all together, we find $(h \circ g)(-1) = h(g(-1)) = h(-3) = 1$.

   (c) To find $(h \circ s)(-2) = h(s(-2))$, we first need $s(-2)$. We see the point $(-2, 3)$ is on the graph of s, so $s(-2) = 3$. Next, we see $(3, -1) \in h$, so $h(3) = -1$. Hence, $(h \circ s)(-2) = h(s(-2)) = h(3) = -1$.

   (d) To find $(f \circ s)(0) = f(s(0))$ we infer from the graph of s that it contains the point $(0, 3)$, so $s(0) = 3$. Sine $f(3) = 6(3) - (3)^2 = 9$, we have $(f \circ s)(0) = f(s(0)) = f(3) = 9$.

2. To find a formula for $(g \circ f)(x) = g(f(x))$, we substitute $f(x) = 6x - x^2$

in for t in the formula for $g(t)$:

$$(g \circ f)(x) = g(f(x)) = g(6x - x^2)$$
$$= \begin{cases} 2(6x - x^2) - 1 & \text{if } -1 \leq 6x - x^2 < 3, \\ (6x - x^2)^2 & \text{if } 6x - x^2 \geq 3. \end{cases}$$

Simplifying each expression, we get $2(6x - x^2) - 1 = -2x^2 + 12x - 1$ for the first piece and $(6x - x^2)^2 = x^4 - 12x^3 + 36x^2$ for the second piece. The real challenge comes in solving the inequalities $-1 \leq 6x - x^2 < 3$ and $6x - x^2 \geq 3$. While we could solve each individually using a sign diagram, a graphical approach works best here. We graph the parabola $y = 6x - x^2$, finding the vertex is $(3, 9)$ with intercepts $(0, 0)$ and $(6, 0)$ along with the horizontal lines $y = -1$ and $y = 3$ below. We determine the intersection points by solving $6x - x^2 = -1$ and $6x - x^2 = 3$. Using the quadratic formula, we find the solutions to each equation are $x = 3 \pm \sqrt{10}$ and $x = 3 \pm \sqrt{6}$, respectively.

From the graph in Figure 5.3.2, we see the parabola $y = 6x - x^2$ is between the lines $y = -1$ and $y = 3$ from $x = 3 - \sqrt{10}$ to $x = 3 - \sqrt{6}$ and again from $x = 3 + \sqrt{6}$ to $x = 3 + \sqrt{10}$. Hence the solution to $-1 \leq 6x - x^2 < 3$ is $[3 - \sqrt{10}, 3 - \sqrt{6}) \cup (3 + \sqrt{6}, 3 + \sqrt{10}]$. We also note $y = 6x - x^2$ is above the line $y = 3$ for all x between $x = 3 - \sqrt{6}$ and $3 + \sqrt{6}$. Hence, the solution to $6x - x^2 \geq 3$ is $[3 - \sqrt{6}, 3 + \sqrt{6}]$. Hence,

$$(g \circ f)(x) = \begin{cases} -2x^2 + 12x - 1 & \text{if } x \in [3 - \sqrt{10}, 3 - \sqrt{6}) \\ & \qquad \cup (3 + \sqrt{6}, 3 + \sqrt{10}], \\ x^4 - 12x^3 + +36x^2 & \text{if } x \in [3 - \sqrt{6}, 3 + \sqrt{6}]. \end{cases}$$

3. Last but not least, we are tasked with representing $s \circ h$ as a set of ordered pairs. Since h is described by the discrete set of points

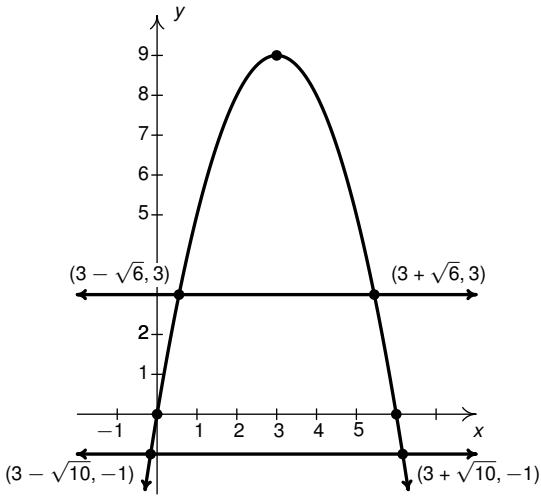
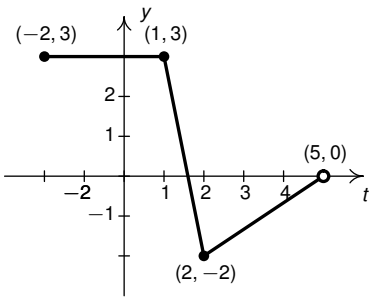


Figure 5.3.2: $y = 6x - x^2$, $y = -1$, $y = 3$

$h = \{(-3,1),(-2,6),(0,-2),(1,5),(3,-1)\}$, we find $s \circ h$ point by point. We keep the graph of s handy and construct the table below to help us organize our work.



$y = s(t)$

x	$h(x)$	$s(h(x))$
-3	1	3
-2	6	undefined
0	-2	3
1	5	undefined
3	-1	3

Since neither 6 nor 5 are in the domain of s, -2 and 1 are not in the domain of $s \circ h$. Hence, we get $s \circ h = \{(-3,3),(0,3),(3,3)\}$. □

A useful skill in Calculus is to be able to take a complicated function and break it down into a composition of easier functions which our last exam-

ple illustrates. As with Example **??**, we want to avoid trivial decompositions, which, when it comes to function composition, are those involving the identity function $I(x) = x$ as described in Theorem 5.3.1.

Example 5.3.3.

1. Write each of the following functions as a composition of two or more (non-identity) functions. Check your answer by performing the function composition.

   (a) $F(x) = |3x - 1|$ (b) $G(t) = \dfrac{2}{t^2 + 1}$ (c) $H(s) = \dfrac{\sqrt{s} + 1}{\sqrt{s} - 1}$

2. For $F(x) = \sqrt{\dfrac{2x - 1}{x^2 + 4}}$, find functions f, g, and h to decompose F nontrivially as $F = f \circ \left(\dfrac{g}{h}\right)$.

Solution. There are many approaches to this kind of problem, and we showcase a different methodology in each of the solutions below.

1. (a) Our goal is to express the function F as $F = g \circ f$ for functions g and f. From Definition 5.3.1, we know $F(x) = g(f(x))$, and we can think of $f(x)$ as being the 'inside' function and g as being the 'outside' function. Looking at $F(x) = |3x - 1|$ from an 'inside versus outside' perspective, we can think of $3x - 1$ being inside the absolute value symbols. Taking this cue, we define $f(x) = 3x - 1$. At this point, we have $F(x) = |f(x)|$. What is the outside function? The function which takes the absolute value of its input, $g(x) = |x|$. Sure enough, this checks: $(g \circ f)(x) = g(f(x)) = |f(x)| = |3x - 1| = F(x)$.

   (b) We attack deconstructing G from an operational approach. Given an input t, the first step is to square t, then add 1, then divide the result into 2. We will assign each of these steps a function so as to write G as a composite of *three* functions: f, g and h. Our first function, f, is the function that squares its input, $f(t) = t^2$. The next function is the function that adds 1

to its input, $g(t) = t + 1$. Our last function takes its input and divides it into 2, $h(t) = \frac{2}{t}$. The claim is that $G = h \circ g \circ f$ which checks:

$$(h \circ g \circ f)(t) = h(g(f(t))) = h(g\left(t^2\right)) = h\left(t^2 + 1\right) = \frac{2}{t^2 + 1} = G(x).$$

(c) If we look $H(s) = \frac{\sqrt{s}+1}{\sqrt{s}-1}$ with an eye towards building a complicated function from simpler functions, we see the expression $\sqrt{s}$ is a simple piece of the larger function. If we define $f(s) = \sqrt{s}$, we have $H(s) = \frac{f(s)+1}{f(s)-1}$. If we want to decompose $H = g \circ f$, then we can glean the formula for $g(s)$ by looking at what is being done to $f(s)$. We take $g(s) = \frac{s+1}{s-1}$, and check below:

$$(g \circ f)(s) = g(f(s)) = \frac{f(s) + 1}{f(s) - 1} = \frac{\sqrt{s} + 1}{\sqrt{s} - 1} = H(s).$$

□

2. To write $F = f \circ \left(\frac{g}{h}\right)$ means

$$F(x) = \sqrt{\frac{2x - 1}{x^2 + 4}} = \left(f \circ \left(\frac{g}{h}\right)\right)(x) = f\left(\left(\frac{g}{h}\right)(x)\right) = f\left(\frac{g(x)}{h(x)}\right).$$

Working from the inside out, we have a rational expression with numerator $g(x)$ and denominator $h(x)$. Looking at the formula for $F(x)$, one choice is $g(x) = 2x - 1$ and $h(x) = x^2 + 4$. Making these identifications, we have

$$F(x) = \sqrt{\frac{2x - 1}{x^2 + 4}} = \sqrt{\frac{g(x)}{h(x)}}.$$

Since F takes the square root of $\frac{g(x)}{h(x)}$, the our last function f is the function that takes the square root of its input, i.e., $f(x) = \sqrt{x}$. We leave it to the reader to check that, indeed, $F = f \circ \left(\frac{g}{h}\right)$. □

We close this section of a real-world application of function composition.

Example 5.3.4. The surface area of a sphere is a function of its radius r and is given by the formula $S(r) = 4\pi r^2$. Suppose the sphere is being inflated so that the radius of the sphere is increasing according to the formula $r(t) = 3t^2$, where t is measured in seconds, $t \geq 0$, and r is measured in inches. Find and interpret $(S \circ r)(t)$.

Solution. If we look at the functions $S(r)$ and $r(t)$ individually, we see the former gives the surface area of a sphere of a given radius while the latter gives the radius at a given time. So, given a specific time, t, we could find the radius at that time, $r(t)$ and feed that into $S(r)$ to find the surface area at that time. From this we see that the surface area S is ultimately a function of time t and we find $(S \circ r)(t) = S(r(t)) = 4\pi(r(t))^2 = 4\pi\left(3t^2\right)^2 = 36\pi t^4$. This formula allows us to compute the surface area directly given the time without going through the 'intermediary variable' r. $\square$

5.3.1 Exercises

In Exercises 1 - 12, use the given pair of functions to find the following values if they exist.

- $(g \circ f)(0)$
- $(f \circ g)(-1)$
- $(f \circ f)(2)$
- $(g \circ f)(-3)$
- $(f \circ g)\left(\frac{1}{2}\right)$
- $(f \circ f)(-2)$

1. $f(x) = x^2$, $g(t) = 2t + 1$
2. $f(x) = 4 - x$, $g(t) = 1 - t^2$
3. $f(x) = 4 - 3x$, $g(t) = |t|$
4. $f(x) = |x - 1|$, $g(t) = t^2 - 5$
5. $f(x) = 4x + 5$, $g(t) = \sqrt{t}$
6. $f(x) = \sqrt{3 - x}$, $g(t) = t^2 + 1$
7. $f(x) = 6 - x - x^2$, $g(t) = t\sqrt{t + 10}$
8. $f(x) = \sqrt[3]{x + 1}$, $g(t) = 4t^2 - t$
9. $f(x) = \dfrac{3}{1 - x}$, $g(t) = \dfrac{4t}{t^2 + 1}$
10. $f(x) = \dfrac{x}{x + 5}$, $g(t) = \dfrac{2}{7 - t^2}$
11. $f(x) = \dfrac{2x}{5 - x^2}$, $g(t) = \sqrt{4t + 1}$
12. $f(x) = \sqrt{2x + 5}$, $g(t) = \dfrac{10t}{t^2 + 1}$

In Exercises 13 - 24, use the given pair of functions to find and simplify expressions for the following functions and state the domain of each using interval notation.

- $(g \circ f)(x)$
- $(f \circ g)(t)$
- $(f \circ f)(x)$

13. $f(x) = 2x + 3$, $g(t) = t^2 - 9$
14. $f(x) = x^2 - x + 1$, $g(t) = 3t - 5$
15. $f(x) = x^2 - 4$, $g(t) = |t|$
16. $f(x) = 3x - 5$, $g(t) = \sqrt{t}$
17. $f(x) = |x + 1|$, $g(t) = \sqrt{t}$
18. $f(x) = 3 - x^2$, $g(t) = \sqrt{t + 1}$
19. $f(x) = |x|$, $g(t) = \sqrt{4 - t}$
20. $f(x) = x^2 - x - 1$, $g(t) = \sqrt{t - 5}$
21. $f(x) = 3x - 1$, $g(t) = \dfrac{1}{t + 3}$
22. $f(x) = \dfrac{3x}{x - 1}$, $g(t) = \dfrac{t}{t - 3}$
23. $f(x) = \dfrac{x}{2x + 1}$, $g(t) = \dfrac{2t + 1}{t}$
24. $f(x) = \dfrac{2x}{x^2 - 4}$, $g(t) = \sqrt{1 - t}$

In Exercises 25 - 30, use $f(x) = -2x$, $g(t) = \sqrt{t}$ and $h(s) = |s|$ to find and simplify expressions for the following functions and state the domain of each using interval notation.

25. $(h \circ g \circ f)(x)$
26. $(h \circ f \circ g)(t)$
27. $(g \circ f \circ h)(s)$
28. $(g \circ h \circ f)(x)$
29. $(f \circ h \circ g)(t)$
30. $(f \circ g \circ h)(s)$

In Exercises 31 - 43, let f be the function defined by

$$f = \{(-3, 4), (-2, 2), (-1, 0), (0, 1), (1, 3), (2, 4), (3, -1)\}$$

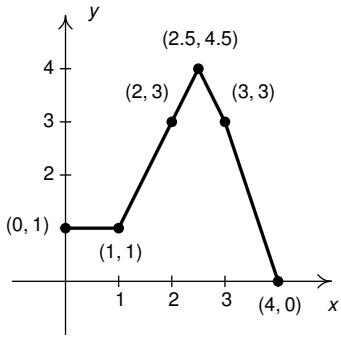
and let g be the function defined by

$$g = \{(-3, -2), (-2, 0), (-1, -4), (0, 0), (1, -3), (2, 1), (3, 2)\}.$$

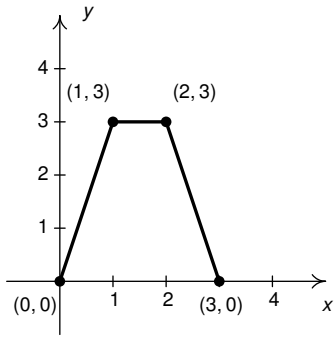
Find the following, if it exists.

31. $(f \circ g)(3)$
32. $f(g(-1))$
33. $(f \circ f)(0)$
34. $(f \circ g)(-3)$
35. $(g \circ f)(3)$
36. $g(f(-3))$
37. $(g \circ g)(-2)$
38. $(g \circ f)(-2)$
39. $g(f(g(0)))$
40. $f(f(f(-1)))$
41. $f(f(f(f(f(1)))))$
42. $\underbrace{(g \circ g \circ \cdots \circ g)}_{n \text{ times}}(0)$
43. Find the domain and range of $f \circ g$ and $g \circ f$.

In Exercises 44 - 50, use the graphs of $y = f(x)$ and $y = g(x)$ below to find the following if it exists.



$y = f(x)$



$y = g(x)$

44. $(g \circ f)(1)$

45. $(f \circ g)(3)$

46. $(g \circ f)(2)$

47. $(f \circ g)(0)$

48. $(f \circ f)(4)$

49. $(g \circ g)(1)$

50. Find the domain and range of $f \circ g$ and $g \circ f$.

In Exercises 51 - 60, write the given function as a composition of two or more non-identity functions. (There are several correct answers, so check your answer using function composition.)

51. $p(x) = (2x + 3)^3$

52. $P(x) = \left(x^2 - x + 1\right)^5$

53. $h(t) = \sqrt{2t - 1}$

54. $H(t) = |7 - 3t|$

55. $r(s) = \dfrac{2}{5s + 1}$

56. $R(s) = \dfrac{7}{s^2 - 1}$

57. $q(z) = \dfrac{|z| + 1}{|z| - 1}$

58. $Q(z) = \dfrac{2z^3 + 1}{z^3 - 1}$

59. $v(x) = \dfrac{2x + 1}{3 - 4x}$

60. $w(x) = \dfrac{x^2}{x^4 + 1}$

61. Write the function $F(x) = \sqrt{\dfrac{x^3 + 6}{x^3 - 9}}$ as a composition of three or more non-identity functions.

62. Let $g(x) = -x$, $h(x) = x + 2$, $j(x) = 3x$ and $k(x) = x - 4$. In what order must these functions be composed with $f(x) = \sqrt{x}$ to create $F(x) = 3\sqrt{-x+2} - 4$?

63. What linear functions could be used to transform $f(x) = x^3$ into $F(x) = -\frac{1}{2}(2x-7)^3 + 1$? What is the proper order of composition?

64. Let $f(x) = 3x+1$ and let $g(x) = \begin{cases} 2x - 1 & \text{if } x \leq 3 \\ 4 - x & \text{if } x > 3 \end{cases}$. Find expressions for $(f \circ g)(x)$ and $(g \circ f)(x)$.

65. The volume V of a cube is a function of its side length x. Let's assume that $x = t + 1$ is also a function of time t, where x is measured in inches and t is measured in minutes. Find a formula for V as a function of t.

66. Suppose a local vendor charges \$2 per hot dog and that the number of hot dogs sold per hour x is given by $x(t) = -4t^2 + 20t + 92$, where t is the number of hours since 10 AM, $0 \leq t \leq 4$.

    (a) Find an expression for the revenue per hour R as a function of x.

    (b) Find and simplify $(R \circ x)(t)$. What does this represent?

    (c) What is the revenue per hour at noon?

67. Discuss with your classmates how 'real-world' processes such as filling out federal income tax forms or computing your final course grade could be viewed as a use of function composition. Find a process for which composition with itself (iteration) makes sense.

5.3.2 Answers

1. For $f(x) = x^2$ and $g(t) = 2t + 1$,
   - $(g \circ f)(0) = 1$
   - $(f \circ f)(2) = 16$
   - $(f \circ g)\left(\frac{1}{2}\right) = 4$
   - $(f \circ g)(-1) = 1$
   - $(g \circ f)(-3) = 19$
   - $(f \circ f)(-2) = 16$
2. For $f(x) = 4 - x$ and $g(t) = 1 - t^2$,
   - $(g \circ f)(0) = -15$
   - $(f \circ f)(2) = 2$
   - $(f \circ g)\left(\frac{1}{2}\right) = \frac{13}{4}$
   - $(f \circ g)(-1) = 4$
   - $(g \circ f)(-3) = -48$
   - $(f \circ f)(-2) = -2$
3. For $f(x) = 4 - 3x$ and $g(t) = |t|$,
   - $(g \circ f)(0) = 4$
   - $(f \circ f)(2) = 10$
   - $(f \circ g)\left(\frac{1}{2}\right) = \frac{5}{2}$
   - $(f \circ g)(-1) = 1$
   - $(g \circ f)(-3) = 13$
   - $(f \circ f)(-2) = -26$
4. For $f(x) = |x - 1|$ and $g(t) = t^2 - 5$,
   - $(g \circ f)(0) = -4$
   - $(f \circ f)(2) = 0$
   - $(f \circ g)\left(\frac{1}{2}\right) = \frac{23}{4}$
   - $(f \circ g)(-1) = 5$
   - $(g \circ f)(-3) = 11$
   - $(f \circ f)(-2) = 2$
5. For $f(x) = 4x + 5$ and $g(t) = \sqrt{t}$,
   - $(g \circ f)(0) = \sqrt{5}$
   - $(f \circ f)(2) = 57$
   - $(f \circ g)\left(\frac{1}{2}\right) = 5 + 2\sqrt{2}$
   - $(f \circ f)(-2) = -7$
   - $(f \circ g)(-1)$ is not real
   - $(g \circ f)(-3)$ is not real
6. For $f(x) = \sqrt{3 - x}$ and $g(t) = t^2 + 1$,
   - $(g \circ f)(0) = 4$
   - $(f \circ f)(2) = \sqrt{2}$
   - $(f \circ g)\left(\frac{1}{2}\right) = \frac{\sqrt{7}}{2}$
   - $(f \circ g)(-1) = 1$
   - $(g \circ f)(-3) = 7$
   - $(f \circ f)(-2) = \sqrt{3 - \sqrt{5}}$
7. For $f(x) = 6 - x - x^2$ and $g(t) = t\sqrt{t + 10}$,
   - $(g \circ f)(0) = 24$
   - $(f \circ f)(2) = 6$
   - $(f \circ g)(-1) = 0$
   - $(g \circ f)(-3) = 0$

- $(f \circ g)\left(\frac{1}{2}\right) = \frac{27-2\sqrt{42}}{8}$
- $(f \circ f)(-2) = -14$

8. For $f(x) = \sqrt[3]{x+1}$ and $g(t) = 4t^2 - t$,

   - $(g \circ f)(0) = 3$
   - $(f \circ f)(2) = \sqrt[3]{\sqrt[3]{3}+1}$
   - $(f \circ g)\left(\frac{1}{2}\right) = \frac{\sqrt[3]{12}}{2}$
   - $(f \circ g)(-1) = \sqrt[3]{6}$
   - $(g \circ f)(-3) = 4\sqrt[3]{4} + \sqrt[3]{2}$
   - $(f \circ f)(-2) = 0$

9. For $f(x) = \frac{3}{1-x}$ and $g(t) = \frac{4t}{t^2+1}$,

   - $(g \circ f)(0) = \frac{6}{5}$
   - $(f \circ f)(2) = \frac{3}{4}$
   - $(f \circ g)\left(\frac{1}{2}\right) = -5$
   - $(f \circ g)(-1) = 1$
   - $(g \circ f)(-3) = \frac{48}{25}$
   - $(f \circ f)(-2)$ is undefined

10. For $f(x) = \frac{x}{x+5}$ and $g(t) = \frac{2}{7-t^2}$,

    - $(g \circ f)(0) = \frac{2}{7}$
    - $(f \circ f)(2) = \frac{2}{37}$
    - $(f \circ g)\left(\frac{1}{2}\right) = \frac{8}{143}$
    - $(f \circ g)(-1) = \frac{1}{16}$
    - $(g \circ f)(-3) = \frac{8}{19}$
    - $(f \circ f)(-2) = -\frac{2}{13}$

11. For $f(x) = \frac{2x}{5-x^2}$ and $g(t) = \sqrt{4t+1}$,

    - $(g \circ f)(0) = 1$
    - $(f \circ f)(2) = -\frac{8}{11}$
    - $(f \circ g)\left(\frac{1}{2}\right) = \sqrt{3}$
    - $(f \circ g)(-1)$ is not real
    - $(g \circ f)(-3) = \sqrt{7}$
    - $(f \circ f)(-2) = \frac{8}{11}$

12. For $f(x) = \sqrt{2x+5}$ and $g(t) = \frac{10t}{t^2+1}$,

    - $(g \circ f)(0) = \frac{5\sqrt{5}}{3}$
    - $(f \circ f)(2) = \sqrt{11}$
    - $(f \circ g)\left(\frac{1}{2}\right) = \sqrt{13}$
    - $(f \circ g)(-1)$ is not real
    - $(g \circ f)(-3)$ is not real
    - $(f \circ f)(-2) = \sqrt{7}$

13. For $f(x) = 2x + 3$ and $g(t) = t^2 - 9$

    - $(g \circ f)(x) = 4x^2 + 12x$, domain: $(-\infty, \infty)$
    - $(f \circ g)(t) = 2t^2 - 15$, domain: $(-\infty, \infty)$
    - $(f \circ f)(x) = 4x + 9$, domain: $(-\infty, \infty)$

14. For $f(x) = x^2 - x + 1$ and $g(t) = 3t - 5$

- $(g \circ f)(x) = 3x^2 - 3x - 2$, domain: $(-\infty, \infty)$
- $(f \circ g)(t) = 9t^2 - 33t + 31$, domain: $(-\infty, \infty)$
- $(f \circ f)(x) = x^4 - 2x^3 + 2x^2 - x + 1$, domain: $(-\infty, \infty)$

15. For $f(x) = x^2 - 4$ and $g(t) = |t|$
    - $(g \circ f)(x) = |x^2 - 4|$, domain: $(-\infty, \infty)$
    - $(f \circ g)(t) = |t|^2 - 4 = t^2 - 4$, domain: $(-\infty, \infty)$
    - $(f \circ f)(x) = x^4 - 8x^2 + 12$, domain: $(-\infty, \infty)$
16. For $f(x) = 3x - 5$ and $g(t) = \sqrt{t}$
    - $(g \circ f)(x) = \sqrt{3x - 5}$, domain: $\left[\frac{5}{3}, \infty\right)$
    - $(f \circ g)(t) = 3\sqrt{t} - 5$, domain: $[0, \infty)$
    - $(f \circ f)(x) = 9x - 20$, domain: $(-\infty, \infty)$
17. For $f(x) = |x + 1|$ and $g(t) = \sqrt{t}$
    - $(g \circ f)(x) = \sqrt{|x + 1|}$, domain: $(-\infty, \infty)$
    - $(f \circ g)(t) = |\sqrt{t} + 1| = \sqrt{t} + 1$, domain: $[0, \infty)$
    - $(f \circ f)(x) = ||x + 1| + 1| = |x + 1| + 1$, domain: $(-\infty, \infty)$
18. For $f(x) = 3 - x^2$ and $g(t) = \sqrt{t + 1}$
    - $(g \circ f)(x) = \sqrt{4 - x^2}$, domain: $[-2, 2]$
    - $(f \circ g)(t) = 2 - t$, domain: $[-1, \infty)$
    - $(f \circ f)(x) = -x^4 + 6x^2 - 6$, domain: $(-\infty, \infty)$
19. For $f(x) = |x|$ and $g(t) = \sqrt{4 - t}$
    - $(g \circ f)(x) = \sqrt{4 - |x|}$, domain: $[-4, 4]$
    - $(f \circ g)(t) = |\sqrt{4 - t}| = \sqrt{4 - t}$, domain: $(-\infty, 4]$
    - $(f \circ f)(x) = ||x|| = |x|$, domain: $(-\infty, \infty)$
20. For $f(x) = x^2 - x - 1$ and $g(t) = \sqrt{t - 5}$
    - $(g \circ f)(x) = \sqrt{x^2 - x - 6}$, domain: $(-\infty, -2] \cup [3, \infty)$

- $(f \circ g)(t) = t - 6 - \sqrt{t-5}$, domain: $[5, \infty)$
- $(f \circ f)(x) = x^4 - 2x^3 - 2x^2 + 3x + 1$, domain: $(-\infty, \infty)$

21. For $f(x) = 3x - 1$ and $g(t) = \frac{1}{t+3}$
    - $(g \circ f)(x) = \frac{1}{3x+2}$, domain: $\left(-\infty, -\frac{2}{3}\right) \cup \left(-\frac{2}{3}, \infty\right)$
    - $(f \circ g)(t) = -\frac{t}{t+3}$, domain: $(-\infty, -3) \cup (-3, \infty)$
    - $(f \circ f)(x) = 9x - 4$, domain: $(-\infty, \infty)$
22. For $f(x) = \frac{3x}{x-1}$ and $g(t) = \frac{t}{t-3}$
    - $(g \circ f)(x) = x$, domain: $(-\infty, 1) \cup (1, \infty)$
    - $(f \circ g)(t) = t$, domain: $(-\infty, 3) \cup (3, \infty)$
    - $(f \circ f)(x) = \frac{9x}{2x+1}$, domain: $\left(-\infty, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, 1\right) \cup (1, \infty)$
23. For $f(x) = \frac{x}{2x+1}$ and $g(t) = \frac{2t+1}{t}$
    - $(g \circ f)(x) = \frac{4x+1}{x}$, domain: $\left(-\infty, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, 0\right), \cup (0, \infty)$
    - $(f \circ g)(t) = \frac{2t+1}{5t+2}$, domain: $\left(-\infty, -\frac{2}{5}\right) \cup \left(-\frac{2}{5}, 0\right) \cup (0, \infty)$
    - $(f \circ f)(x) = \frac{x}{4x+1}$, domain: $\left(-\infty, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, -\frac{1}{4}\right) \cup \left(-\frac{1}{4}, \infty\right)$
24. For $f(x) = \frac{2x}{x^2-4}$ and $g(t) = \sqrt{1-t}$
    - $(g \circ f)(x) = \sqrt{\frac{x^2-2x-4}{x^2-4}}$, domain: $(-\infty, -2) \cup \left[1 - \sqrt{5}, 2\right) \cup \left[1 + \sqrt{5}, \infty\right)$
    - $(f \circ g)(t) = -\frac{2\sqrt{1-t}}{t+3}$, domain: $(-\infty, -3) \cup (-3, 1]$
    - $(f \circ f)(x) = \frac{4x-x^3}{x^4-9x^2+16}$, domain: $\left(-\infty, -\frac{1+\sqrt{17}}{2}\right) \cup \left(-\frac{1+\sqrt{17}}{2}, -2\right) \cup \left(-2, \frac{1-\sqrt{17}}{2}\right) \cup \left(\frac{1-\sqrt{17}}{2}, \frac{-1+\sqrt{17}}{2}\right) \cup \left(\frac{-1+\sqrt{17}}{2}, 2\right) \cup \left(2, \frac{1+\sqrt{17}}{2}\right) \cup \left(\frac{1+\sqrt{17}}{2}, \infty\right)$
25. $(h \circ g \circ f)(x) = \left|\sqrt{-2x}\right| = \sqrt{-2x}$, domain: $(-\infty, 0]$
26. $(h \circ f \circ g)(t) = \left| -2\sqrt{t}\right| = 2\sqrt{t}$, domain: $[0, \infty)$
27. $(g \circ f \circ h)(s) = \sqrt{-2|s|}$, domain: $\{0\}$

28. $(g \circ h \circ f)(x) = \sqrt{|-2x|} = \sqrt{2|x|}$, domain: $(-\infty, \infty)$
29. $(f \circ h \circ g)(t) = -2|\sqrt{t}| = -2\sqrt{t}$, domain: $[0, \infty)$
30. $(f \circ g \circ h)(s) = -2\sqrt{|s|}$, , domain: $(-\infty, \infty)$
31. $(f \circ g)(3) = f(g(3)) = f(2) = 4$
32. $f(g(-1)) = f(-4)$ which is undefined
33. $(f \circ f)(0) = f(f(0)) = f(1) = 3$
34. $(f \circ g)(-3) = f(g(-3)) = f(-2) = 2$
35. $(g \circ f)(3) = g(f(3)) = g(-1) = -4$
36. $g(f(-3)) = g(4)$ which is undefined
37. $(g \circ g)(-2) = g(g(-2)) = g(0) = 0$
38. $(g \circ f)(-2) = g(f(-2)) = g(2) = 1$
39. $g(f(g(0))) = g(f(0)) = g(1) = -3$
40. $f(f(f(-1))) = f(f(0)) = f(1) = 3$
41. $f(f(f(f(f(1))))) = f(f(f(f(3)))) = f(f(f(-1))) = f(f(0)) = f(1) = 3$
42. $\underbrace{(g \circ g \circ \cdots \circ g)}_{n \text{ times}}(0) = 0$
43.
    - The domain of $f \circ g$ is $\{-3, -2, 0, 1, 2, 3\}$ and the range of $f \circ g$ is $\{1, 2, 3, 4\}$.
    - The domain of $g \circ f$ is $\{-2, -1, 0, 1, 3\}$ and the range of $g \circ f$ is $\{-4, -3, 0, 1, 2\}$.
44. $(g \circ f)(1) = 3$
45. $(f \circ g)(3) = 1$
46. $(g \circ f)(2) = 0$
47. $(f \circ g)(0) = 1$
48. $(f \circ f)(4) = 1$
49. $(g \circ g)(1) = 0$
50.
    - The domain of $f \circ g$ is $[0, 3]$ and the range of $f \circ g$ is $[1, 4.5]$.
    - The domain of $g \circ f$ is $[0, 2] \cup [3, 4]$ and the range is $[0, 3]$.

51. Let $f(x) = 2x + 3$ and $g(x) = x^3$, then $p(x) = (g \circ f)(x)$.

52. Let $f(x) = x^2 - x + 1$ and $g(x) = x^5$, $P(x) = (g \circ f)(x)$.

53. Let $f(t) = 2t - 1$ and $g(t) = \sqrt{t}$, then $h(t) = (g \circ f)(t)$.

54. Let $f(t) = 7 - 3t$ and $g(t) = |t|$, then $H(t) = (g \circ f)(t)$.

55. Let $f(s) = 5s + 1$ and $g(s) = \frac{2}{s}$, then $r(s) = (g \circ f)(s)$.

56. Let $f(s) = s^2 - 1$ and $g(s) = \frac{7}{s}$, then $R(s) = (g \circ f)(s)$.

57. Let $f(z) = |z|$ and $g(z) = \frac{z+1}{z-1}$, then $q(z) = (g \circ f)(z)$.

58. Let $f(z) = z^3$ and $g(z) = \frac{2z+1}{z-1}$, then $Q(z) = (g \circ f)(z)$.

59. Let $f(x) = 2x$ and $g(x) = \frac{x+1}{3-2x}$, then $v(x) = (g \circ f)(x)$.

60. Let $f(x) = x^2$ and $g(x) = \frac{x}{x^2+1}$, then $w(x) = (g \circ f)(x)$.

61. $F(x) = \sqrt{\frac{x^3+6}{x^3-9}} = (h(g(f(x)))$ where $f(x) = x^3$, $g(x) = \frac{x+6}{x-9}$ and $h(x) = \sqrt{x}$.

62. $F(x) = 3\sqrt{-x+2} - 4 = k(j(f(h(g(x)))))$

63. One solution is $F(x) = -\frac{1}{2}(2x - 7)^3 + 1 = k(j(f(h(g(x)))))$ where $g(x) = 2x$, $h(x) = x - 7$, $j(x) = -\frac{1}{2}x$ and $k(x) = x + 1$. You could also have $F(x) = H(f(G(x)))$ where $G(x) = 2x - 7$ and $H(x) = -\frac{1}{2}x + 1$.

64. $(f \circ g)(x) = \begin{cases} 6x - 2 & \text{if } x \leq 3 \\ 13 - 3x & \text{if } x > 3 \end{cases}$ and $(g \circ f)(x) = \begin{cases} 6x + 1 & \text{if } x \leq \dfrac{2}{3} \\ 3 - 3x & \text{if } x > \dfrac{2}{3} \end{cases}$

65. $V(x) = x^3$ so $V(x(t)) = (t + 1)^3$

66. (a) $R(x) = 2x$

    (b) $(R \circ x)(t) = -8t^2 + 40t + 184$, $0 \leq t \leq 4$. This gives the revenue per hour as a function of time.

    (c) Noon corresponds to $t = 2$, so $(R \circ x)(2) = 232$. The hourly revenue at noon is \$232 per hour.

5.4 Transformations of Graphs

5.5 Relations and Implicit Functions

5.6 Inverse Functions